

Fig. 2: Actual-size PCB layout of the DC motor soft-starter

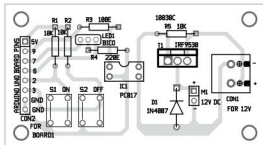


Fig. 3: Components layout for the PCB

gives additional protection over induced voltage when the motor is turned off.

To Board1, enclose the assembled PCB along with Board1 in a suitable plastic box.

The circuit's operation is controlled using the software program loaded into the internal memory of Arduino Uno board. The program (softs.ino) is written in Arduino programming language sketch. Arduino IDE is used to compile and upload the program to the Arduino board.

### Construction and testing

An actual-size PCB layout of the DC motor soft-starter is shown in Fig. 2 and its components layout in Fig. 3. After assembling the circuit on a PCB, connect CON2 to Arduino Uno board through external male-to-male jumper wires.

### EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at [source.efymag.com](http://source.efymag.com)

The circuit works off the 5V USB power supply used for Arduino Uno board. Regulated 12V power supply is used to operate the DC motor.

### Further application

The circuit can be converted into an AC motor soft-starter by using thyristors in place of MOSFET T1. It can be used along with device control projects. Additional protection and closed-loop control can be included, if required. Different timings and PWM values can be set for both soft-start and soft-stop. **EFY**



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